

TAMIL NADU STATE BOARD - STANDARD 10 MATHEMATICS

CHAPTER 7: MENSURATION

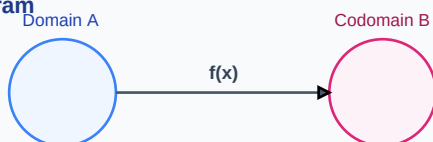
PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 10 English Medium
Subject:	Mathematics
Chapter Name:	Mensuration

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Function Mapping Diagram



Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Surface Area	The total area of the outer faces of a three-dimensional solid, split into lateral (LSA) and total (TSA) areas.
Volume	The amount of space occupied by a three-dimensional object, measured in cubic units.
Cylinder	A solid geometrical figure with straight parallel sides and a circular cross-section.
Frustum of Cone	The portion of a cone that lies between the base and a plane parallel to the base cutting off the top.

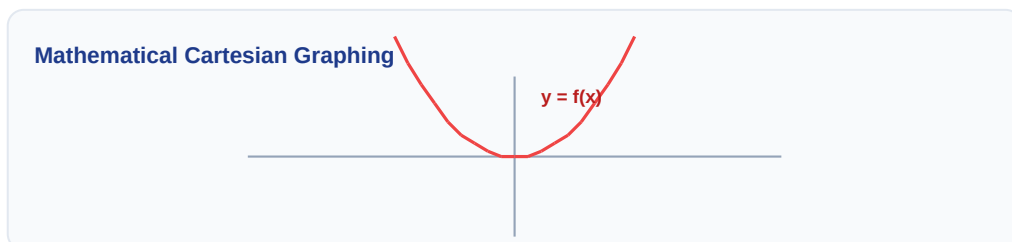
Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

Venn Diagram: $A \cap B$



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:



Essential Formulas, Laws & Identities:

- Cylinder: $CSA = 2 * \pi * r * h$ | $TSA = 2 * \pi * r * (r + h)$
- Cone: $CSA = \pi * r * l$ | $TSA = \pi * r * (r + l)$ where slant height $l = \sqrt{r^2 + h^2}$
- Sphere: Surface Area = $4 * \pi * r^2$ | Volume = $(4/3) * \pi * r^3$
- Hemisphere: $CSA = 2 * \pi * r^2$ | $TSA = 3 * \pi * r^2$ | Volume = $(2/3) * \pi * r^3$
- Frustum of Cone: Volume = $(1/3) * \pi * h * (R^2 + r^2 + R*r)$

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: Calculate the volume of a cylinder of radius 7 cm and height 10 cm. (Take $\pi = 22/7$)

Step-by-step Solution:

Apply volume formula: $V = \pi * r^2 * h$.

$V = (22/7) * 7 * 7 * 10 = 22 * 7 * 10 = 1540$ cubic centimeters (cm^3).

Linear Algebraic Matrix Representation

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$$

HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: A metallic sphere of radius 4.2 cm is melted and recast into the shape of a cylinder of radius 6 cm. Find the height of the cylinder.

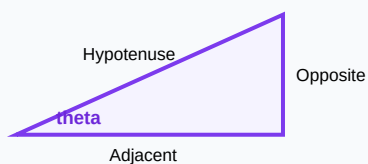
Analytical Solution Strategy:

Volume of sphere = Volume of cylinder.

$$\left(\frac{4}{3}\right) * \pi * r_s^3 = \pi * r_c^2 * h.$$

$$\left(\frac{4}{3}\right) * (4.2)^3 = 6^2 * h \Rightarrow \left(\frac{4}{3}\right) * 74.088 = 36 * h \Rightarrow 98.784 = 36h \Rightarrow h = 2.744 \text{ cm}.$$

Right-Angled Triangle Geometry



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): Find the slant height of a cone of radius 6 cm and height 8 cm.

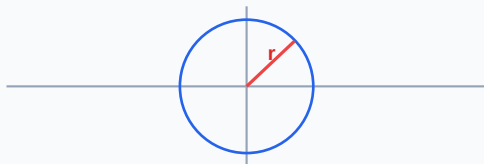
Q2 (2-Mark): Determine the surface area of a sphere of radius 14 cm.

Q3 (2-Mark): Find the TSA of a solid hemisphere of radius 7 cm.

Q4 (5-Mark): A bucket in the shape of a frustum has radii 20 cm and 8 cm and height 16 cm. Find its volume.

Q5 (5-Mark): How many spherical lead shots of radius 2cm can be made from a solid cuboid of dimensions 22cm x 18cm x 16cm?

Circle with Radius Vector on Plane



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Industrial Container Design

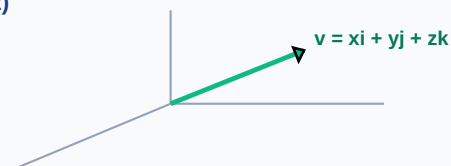
Mensuration formulas calculate volume capacities of chemical containers, fuel tankers, and shipping crates.

Application Case: Material Recasting

Metal foundries utilize volume conservation rules to melt bulk ores and recast them into engine components.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

3D Vector Projections (x, y, z)



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

Probability Tree Outcome Diagram

